

Group 5 - Project Report

How Extreme Can Cyber Attacks Become?

Modeling Yearly Maximums and Future Cyber Risk

Måns Conradson & Jacob Telander

November 2025

Abstract

This report investigates cyber security breaches from the years 2004 - 2022 using extreme value theory (EVT). Annual maximum data breach sizes are analyzed using a block maximum approach and modeled with Generalized Extreme Value (GEV) distributions. Both stationary and non-stationary models are estimated and compared using maximum likelihood, information criteria, and likelihood ratio tests. The results indicate heavy-tailed behavior in extreme data breaches, implying that future incidents could exceed historically observed magnitudes. Estimated return levels suggest that breaches involving several billion compromised records are plausible for long return periods such as 50 or 100 years. However, no statistically significant temporal trend in the magnitude of annual maximum breaches is found, and model comparison favors a stationary specification. Bootstrap analysis and simulation studies reveal substantial uncertainty in long-term predictions, highlighting the limitations of extreme value modeling with short time series. Overall, the findings emphasize both the usefulness and the challenges of applying EVT to cyber risk assessment.

Contents

1	Introduction	3
2	Data	4
3	Methods	6
3.1	Short on Extreme Value Theory (EVT)	6
3.2	Simulation Study	8
3.3	Short on Model Performance and Comparison	8
3.4	Implementation	9
4	Results	10
4.1	Explanatory analysis	10
4.2	Inferential Analysis	13
4.2.1	Stationary Model	13
4.2.2	Non-Stationary Model	14
4.2.3	Model comparison	15
4.2.4	Uncertainty Assessment	15
4.2.5	Simulation Study Results	15
5	Discussion and Conclusion	17
6	Appendix	20
6.1	Model Diagnostics	20
6.2	Additional exploratory plots	21

1 Introduction

With the present digital era, computers and screens occupy an increasingly large part of society, with more and more information being stored digitally. As cyber security incidents become more frequent, protecting computer systems, networks and data from unauthorized damage or illegal extraction has never been more relevant than today. Understanding these breaches is important for organizations and companies that want to evaluate risk and prepare for potential future incidents. One way of doing this is through EVT.

Extreme value analysis (or EVT) is a statistical technique that focuses directly on the most extreme outcomes, rather than average numbers in a data set. The main objective of EVT is to evaluate or predict the probabilities of events that are rarely observed. When it comes to cyber security, larger breaches have larger repair costs and institutional damage, making these incidents more important and of greater interest to analyze and understand. Although EVT is used mainly for subjects such as finance and meteorology, it has been applied to cybersecurity before. For example, Zhan et al (2015) used extreme value analysis to predict cyberattack rates, where they concluded that EVT could be used for long-term prediction.

The project aims to analyze cybersecurity online to see how it has evolved over time. This will be done with the help of extreme value analysis applied on a data set consisting of observed security breaches online that span from 2004 to 2024. To guide the analysis, the project will address the following research questions.

How large can future extreme data breaches realistically become?

Has the magnitude of the largest annual data breaches increased over time, or is it suggesting a trend?

How uncertain are predictions of rare cyber events, such as breaches expected to occur once every 50 or 100 years?

Does a non-stationary extreme value model (allowing trends over time) provide a better description of the data than a stationary model?

2 Data

The data used in the report are retrieved from Kaggle.com and consist of a collection of some of the largest data breaches in history. The data set, named “*World’s Biggest Data Breaches and Hacks*”, contains 419 observations and spans between 2004 and 2022. However, not all observations are used in the analysis, as a block maxima approach is applied where only the largest data breach from each year is retained. The analysis therefore focuses on annual maximum breach sizes, which form the basis for the extreme value analysis.

There are several descriptive variables available in the dataset, however, to concentrate the analysis and align it with the chosen statistical method, two variables are selected. The reduced set of variables and their respective descriptions are presented in Table 1. Figure 1 displays the trend in the size of the largest breach each year, as well as showing how many extreme hacks are completed each year.

Variable	Description
year	The calender year in which the breach or hack occurred. This variable is used to define annual blocks and to investigate potential temporal trends in extreme breach sizes.
records_lost	The number of records lost when the breach or hack occurred. It is measured in amounts of accounts and data records exposed.

Table 1: Variables used in the report

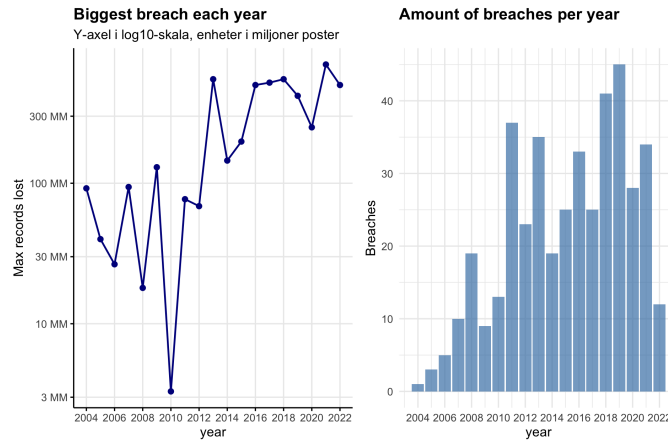


Figure 1: Largest Breaches and Amount of Breaches Made Each Year 2004-2022

In figure 1, the observed number of breaches that are made each year have generally increased since the beginning of the time series. While some years experience a decrease in extreme cyber security hacks compared to previous years, the general trend is increasing with a peak between 2018-2019, although a drastic decrease in the number of completed hacks can be observed in the last year of the series. The largest breach each year follows a quite choppy pattern with a more steady, increased size from year 2013 and onward. The peak is observed in 2021, whilst the largest hack completed in 2010 is the lowest of all maximums in the whole period.

To create block maximums, an observation is chosen for each year. The observation chosen will be the largest reported breach, and thus follow the mathematical expression as:

$$X_t = \max(\text{records_lost year } t)$$

After obtaining the data and loading them into R, the data set is cleaned using the “dplyr” package. This allows for selection of the desired variables for the analysis, removal of any missing values, grouping of observations, and lastly extraction of the maximum values each year, ensuring numerical consistency and suitability for extreme value modeling.

Specifically, the variable “records_lost” contains non-numeric characters, such as text labels and missing values. These entries are removed before all remaining values are converted to numeric form. Furthermore, for the exploratory analysis, a logarithmic transformation of the breach sizes is applied to better visualize the proportions and nature of the data, as the heavy tails in the data make many observations cluster together. The base-10 logarithm of the number of lost records is used in the plots to reduce skewness and improve interoperability. The transformations are solely used for visualization and do not affect any of the estimations in the extreme value models.

When exploring the reliability of the data set, multiple aspects are considered. Overall, the dataset is concluded to be relatively reliable because of its use of well-known events reported over multiple news platforms such as NY Times, BBC and more, as well as official statements from involved companies. In addition, the authors and publishers of the dataset are publicly shown on multiple platforms and provide contact information increasing transparency.

Nevertheless, several limitations of the data set are present. First of all, cases of data-breaches might be under-reported, especially for earlier years and for smaller organizations, but also as a consequence of companies wanting to hide high-profile incidents (Lydon, 2021). Secondly, larger breaches can often be observed to be rounded to the closest million, for example “500 million accounts”, leading to measurement uncertainties. Thirdly, as the observations are fetched from multiple sites and news platforms, definitions might vary slightly when it comes to the variable “records lost”. Lastly, the uncertainty in the estimation of

parameters is increased drastically when the number of observations is reduced, a natural limitation of extreme value analysis.

3 Methods

The methods section begins with explaining the basic concepts of extreme value theory. This is followed by an explanation of the simulation study which is conducted in the report. Model performance measurement and estimation techniques are then discussed with the segment ending with a review of how theory and data are implemented into the analysis.

3.1 Short on Extreme Value Theory (EVT)

In contrast to many classical statistical methods, EVT is specifically developed to focus on the tails of a set of data and the most extreme observations (Coles, 2001). Data breaches in particular are characterized by a highly skewed distribution, where a small number of incidents account for a disproportionately large share of the total number of compromised records. Classical models that focus on average behavior thus fail to capture the risk associated with rare but catastrophic events. Extreme value theory is particularly well suited for this type of problem, as it provides a framework for modeling behaviors of the largest observations in the data.

Two main model strategies are often used within EVT, the block maxima method and the peaks over threshold method. When using block maxima, observations are grouped into blocks, or segments of same lengths, where only the largest observation in each segment is kept. Thus, the data set is often greatly reduced, and most observations are left out of the analysis. The peaks over threshold (POT) method instead uses the implementation of a threshold which is chosen by the analyst at some fitting level. All spikes exceeding the fitted threshold are then implemented in the analysis, meaning that more observations can be used. The choice of threshold often causes the problem of balancing bias and variance in the data. Choosing too low thresholds may mean that important theoretical assumptions are broken whilst a threshold that is too high gives large uncertainty (Coles, 2001).

Whilst both methods have advantages and disadvantages, this report focuses on using block maximums, taking the largest data breach of each year in the data, with the aim of describing the following behavior:

$$M_n = \max\{X_1, \dots, X_n\}$$

M_n represents the maximum value observed in a time series over time n and $X_1, \dots, X_n$ represents a sequence of independent random variables with a common distribution (Coles, 2001). The limiting distribution of M_n can be determined and given through what is called the extremal types theorem, which forms much of the theoretical basis in the subject.

The extremal types theorem is an analogy to the central limit theorem, which states that a rescaled (normalized) sample maxima converges in distribution to one of three extreme value families of distributions. These are known as the Gumbel, Fréchet and Weibull families (Coles, 2001). Early usage of extreme value theory often tried to approximate a found distribution towards one of these three, however, problems regarding this technique firstly arises as there has to be a well defined approach to choosing distribution. Secondly, once this is determined, the unreliability that arises from assuming right choice is made can not be accounted for. Therefore, the three distributions can be combined to form the generalized extreme value (GEV) family of distributions which simplifies its implementation. The GEV distribution function model consists of three parameters and looks as follows:

$$G(z) = \exp\{-[1 + \xi(\frac{z - \mu}{\sigma})]^{-1/\xi}\}$$

The model is defined when $1 + \xi(z - \mu)/\sigma > 0$. Furthermore, μ is the location parameter and can range from $-\infty < \mu < \infty$. σ is a scale parameter which must satisfy $\sigma > 0$ and lastly the shape parameter ξ can range between $-\infty < \xi < \infty$. All three parameters are estimated using maximum likelihood methods, which according to Coles is the preferred method of inference when it comes to EVT (2001). Additionally, the shape parameter ξ is crucial to the modeling technique as it determines the behavior in the tail of the distribution. Different values of ξ implies different important aspects of a distribution as when positive, it suggests there is a heavily-tail in the data, with plausible extreme observations far out in the tail. When $\xi = 0$, the distribution equals a special case where it follows the gumbel family explicitly and when the parameter is negative it implies there is a finite upper bound (Coles, 2001).

An important objective when fitting GEV models to a set of data is to estimate return levels, which describe how large an event is expected to be exceeded on average once during a given time period. For example, a 50 year event represents the size of a data breach that is expected to be exceeded once every 50 years. The return levels of these events provide an interpretable measure of risk and are particularly useful for decision-making and risk assessment.

The block maxima approach relies on the assumption that annual maximums are independent and identically distributed. Although this assumption simplifies modeling, it may be violated in practice due to changes in reporting practices or structural changes in cyber security over time. Consequently, all inferences should be interpreted with caution, particularly when forecasting longer return periods. Furthermore, the implementation of annual maximums means that the report only deals with 19 observations in total. It is well known that the block maximum method leads to great uncertainty, in particular when dealing with few observations. Thus, to explore that uncertainty, a simulation study is also conducted to help determine bias and variance levels, putting the empirical findings in a new perspective.

3.2 Simulation Study

A simulation in the report is conducted to assess performance of the extreme value models under controlled conditions. It is constructed by generating synthetic annual maximums from a GEV distribution. In the simulation, the scale and shape parameters are held constant, whilst the location parameter is given a linear trend, meaning the simulation is designed to follow a scenario where the magnitude of the extreme data breaches increases over time. For each simulated dataset, both a stationary GEV model and a non-stationary GEV model with a time-dependent location parameter are fitted using maximum likelihood estimation. This process is repeated 2000 times to examine the behavior of the estimated return levels under controlled conditions.

The purpose of the simulation is not to replicate the real data-generating process exactly, but to investigate whether the estimation procedure behaves as expected when the true underlying process is known. In particular, the simulation allows for a comparison of bias and uncertainty between stationary and non-stationary models when a temporal trend is present.

3.3 Short on Model Performance and Comparison

The performance of the stationary and the non-stationary model is evaluated using the akaike's information criterion (AIC). The measurement weights model complexity against model fit and provides information about relative performance (Burnham and Anderson 2003). The AIC value is computed as:

$$AIC = -2\ln(L) + 2k$$

In the expression, $\ln(L)$ represents the logarithmic likelihood function of the model and k is the number of parameters in the model (Burnham and Anderson 2003). Lower AIC values indicate better relative performance overall, and to formally test significantly different performance levels between models, a likelihood ratio test in addition to diagnostic plots are also applied.

The likelihood ratio test is used to compare two nested models to see if an increase in complexity for a model leads to a statistically significant increase in model performance. The test follows the Chi-square distribution and, for the case of this study, is used to evaluate whether a trend in the location parameter of a GEV model provides a significantly better model fit than when a trend is not present.

3.4 Implementation

The statistical analysis is implemented in the statistical application "R" using the "extRemes" package. The package is specifically designed for extreme value analysis and provides tools for fitting Generalized Extreme Value (GEV) models, calculating return levels, and performing diagnostic checks. Moreover, it is particularly well suited for block maxima approaches and allows for both stationary and non-stationary model specifications, making it appropriate for modeling extreme data breaches over time (Gilleland and Katz, 2016).

The main function used for model estimation is `fevd()`. The function fits a GEV model using maximum likelihood estimation. Utilizing the function, the user is allowed to change all parameter values in the model. For this analysis, this means the location parameter specifically is specified in regards to time whilst all other parameters are held constant, enabling a direct comparison between stationary models and models that incorporate temporal trends. Once a model is fitted, the return levels from specific return periods are calculated using the `return.levels()` function.

For visualization of the results, the package "ggplot2" is primarily used. The package allows the user to create tidy plots and diagrams in an easily understood manner, helping the reader through the report.

After obtaining the data and having it read into R, the report also utilized the "dplyr" package. This allows for selection of the desired variables for the analysis, removal of any missing values, grouping of observations, and lastly extraction of the maximum values each year, setting up the foundation for an extreme value analysis to be performed.

To assess whether the estimation procedure produces reasonable results, the simulation study is performed. The "evd" package is used for data generation in the simulation study, specifically through the utilization of the `rgev()` function. The function allows synthetic data to be generated from a GEV distribution with known parameters.

Other packages are also used in the analysis, however, all statistical inference related to extreme values was carried out using "extRemes". As a result, the choice of supporting packages does not influence the substantive results but rather aids implementation in "R".

4 Results

This section goes through the results of the analysis. Further statistical analysis is firstly presented explaining the nature of the data which is analyzed, providing evidence for the usage of extreme value theory. The results of fitting extreme value models to the data is thereafter presented with parameter and return level estimations, as well as model diagnostics. Bootstrap estimations for the uncertainty in the analysis is then discussed whilst a simulation study ends the section.

4.1 Explanatory analysis

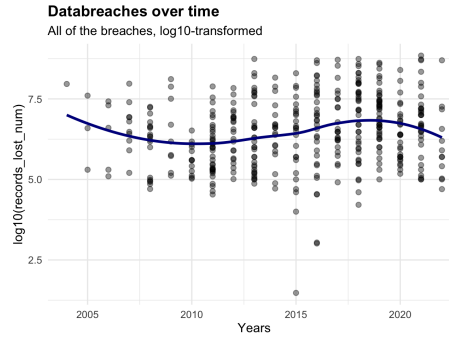


Figure 2: Time series of extreme data breaches

Figure 2 displays a scatterplot of every observed data breach in the original data set. The plot provides an overview of how breach sizes are distributed over time and displays temporal trends that might influence the analysis. A regression line is included to visualize the overall trend.

It shows that the amounts of extreme leaks observed each year has had a steady increase when comparing early years with more recent ones. The regression line furthermore shows that breach severity increased approximately between years 2011 - 2019, with general sizes of breaches declining thereafter, from around 2020 onward. A plausible explanation for the observed relationship may stem from an increase in conducted breaches with smaller severity, resulting in an increase in amount of leaks but not as many data records exposed. More interestingly however, the plot showcases the presence of rare but exponentially large breaches, indicating analysis of average behaviors would fail to capture the risk of interest.

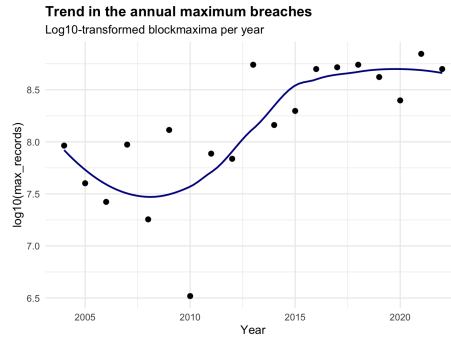


Figure 3: Annual maxima with regression line

Figure 3 displays a scatterplot focusing solely on the most extreme observed data breach each year from 2004-2022. The plot shows that the overall trend in the most extreme data leak has increased from 2006 onward with the most recent observations leveling off. This view of the data is directly aligned with the modeling which is performed in subsequent analysis and although limited observations are displayed in the figure, the pattern clearly shows an evolution in extreme events over time. Moreover, the plot showcase a distinctively different evolution from figure 2, indicating that extreme events can and should be modeled separately.

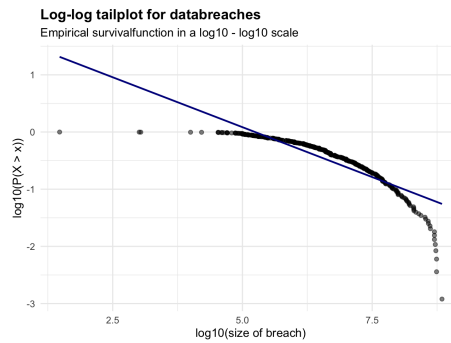


Figure 4: Log-log tailplot for data breaches

Figure 4 showcases the empirical survival function of the data utilizing a log-log tailplot. The figure is used to assess tail behavior in the data and does so by displaying the probability of different data breaches occurring depending on size. In such a plot, approximate linearity over a range of large observations indicates heavy-tailed behavior consistent with extreme value theory. Curvature in the very upper tail typically reflects increased sampling variability due to the scarcity of extreme observations and should not be overinterpreted. The figure

shows that the bulk of the total larger observations approximately follows a linear trend, suggesting there is heavy tail behavior in the upper tail of the distribution. Whilst strong deviations from linearity is observed for smaller breach sizes, they do not describe the bulk of the distribution, and are of little interest to the extreme value modeling in this report. The main behavior does thus provide empirical evidence for the usage of extreme value theory.

Overall, the exploratory analysis reveals several key characteristics of the data breach sizes that motivate the use of extreme value theory. The data are highly skewed, with rare but severe events present in the upper tail. The magnitude of extreme breaches exhibits some temporal structure and has tail behavior consistent with heavy-tailed distributions. These characteristics indicate that classical statistical models centered on average behavior are inadequate for capturing the risk posed by extreme data breaches. Consequently, extreme value theory provides a well-motivated and appropriate framework for modeling the annual maximum breach sizes.

4.2 Inferential Analysis

4.2.1 Stationary Model

For the GEV estimation, modeling is evaluated using maximum likelihood applied to the block maximums each year. The parameter values for the stationary model is presented in table 2

Location	Scale	Shape
124.255	137.289	0.517

Table 2: Estimated parameters for stationary GEV

The table highlights a positive shape parameter ($\hat{\xi} = 0.517$), suggesting there is heavy tail behavior present. This would imply that extreme events, exceeding those that have been recorded historically, could arise. The scale parameter is relatively large, reflecting that there is substantial variability in the annual extremes.

Model diagnostics for the stationary model are found in appendix A. From observing figure x, the QQ-plots shows that the model fits reasonably well with some deviations in the right tail. The density plot reveals that the model has difficulty capturing the shape of the bulk of the distribution and also having trouble with the rightmost tail. The return period plot shows an increase in confidence bounding with a growing return period, which indicates a growing uncertainty when evaluating extremes. Overall, no severe violations of model assumptions are unveiled. Large uncertainty is however evident in the upper tail, an observations which is not unexpected given the limited number of block maximums.

50-year	100-year
1857.020	2726.505

Table 3: Stationary GEV return levels for 50- and 100-year event (million records lost)

Based on the stationary model, the estimated 50-year return level is approximately 1 857 million records, while the estimated 100-year return level was approximately 2 727 million records. These values represent the size of data breaches expected to be exceeded, on average, once every 50 and 100 years, respectively.

4.2.2 Non-Stationary Model

The non-stationary model is implemented to investigate potential temporal trends, analyzing if a model capturing this behavior performs better than a simple stationary model.

μ_0	μ_1	Scale	Shape
144.131	-0.019	118.049	0.625

Table 4: Estimated parameters for non-stationary GEV

The parameter values for the non-stationary model can be found in table 3. Replacing the location parameter is μ_0 and μ_1 which is the location parameter over time, where μ_0 is the baseline value when the time is at zero and should not be interpreted alone, and μ_1 equals the coefficient or slope of the time trend. μ_1 is shown to be a very small negative value, indicating no systematic trend of increased magnitude in extreme data breaches. Whilst not presented in the report, the associated uncertainty of said parameter was also large, further supporting trends might not be present in the data. The shape parameter remains positive and of similar magnitude to that of the stationary model, suggesting that the heavy-tailed nature of the data is robust to the inclusion of a trend.

Model diagnostics for the non-stationary GEV model are furthermore found in appendix A. From figure x in appendix A, the observed QQ-plot show a good fit with fewer deviations in the rightmost tail. This suggests an improved fit compared to the stationary model. The density plot shows inconsistency in the bulk of the distribution and in the rightmost tail, although the general fit seems to improve. The index plot shows that the return levels are year dependent.

Estimation year	50-year	100-year
2004	2079.628	3262.565
2022	2079.281	3262.217

Table 5: Non-stationary GEV return levels

For the non-stationary GEV model, return levels depend on the year at which they are evaluated through the time-varying location parameter. Return levels were therefore computed for different years in the dataset to illustrate the effect of the estimated temporal trend. Table 5 presents the estimated 50- and 100-year return levels evaluated at the first and last years in the sample period. The resulting values are very similar, reflecting the fact that the estimated trend in the location parameter is small and statistically insignificant.

4.2.3 Model comparison

When comparing the models, both an AIC comparison and a likelihood ratio test are used. The AIC value indicated that the stationary model is preferred over a non-stationary, having a value of 260.193 compared to 261.918 in the non-stationary models.

Utilizing a likelihood ratio test using 0.05 as the alpha, much of the same conclusion can be drawn. The tested produced a non-significant result with a p-value around 0.60. Therefore, the null-hypothesis, which states that the stationary GEV model is sufficient, is not rejected. All and all these findings does not suggest that including temporal trends in the modeling significantly improves model fit.

4.2.4 Uncertainty Assessment

To further evaluate model performance and the uncertainty in estimations a non-parametric bootstrap procedure is applied to the 100-year-level returns of the stationary model, seemingly being the preferred design for model fit. The bootstrap quintile results are found in table 6.

	2.5%	50%	97.5%
Normal bootstrap	547.591	2259.633	52164.235
Trimmed bootstrap	547.588	2251.488	30325.500

Table 6: Bootstrap quantiles

From table 6, the bootstrap sampling has a median of 2259.633, while the 95% confidence interval reaches from 547.591 to 52164.235. Approximately 16% of bootstrap replications resulted in numerically unstable fits, a phenomenon which is a quite well known issue when modeling heavy-tailed data with small sample sizes. Constructing a 95% confidence interval for the estimate resulted in a extremely large range, indicating substantial uncertainty in the upper tail. As a robustness check, trimming of the largest 0.5% of bootstrap estimates is conducted. The trimming of the bootstrap reduced the upper bound to approximately 30 325 million records, with the median essentially being unchanged. This result would imply that central tendencies of the return level estimates are relatively stable, whereas the upper uncertainty is driven by a small number of extreme bootstrap estimates. The intervals indicate that there is large uncertainty in estimations of extreme events for data breaches given the sample size, whilst heavy-tail behavior once again is acknowledged when conducted a bootstrap sampling procedure.

4.2.5 Simulation Study Results

A simulation study was conducted to evaluate the performance of stationary and non-stationary GEV models under controlled conditions where the true

data-generating process is known. Annual block maxima were generated from a GEV distribution with a linear trend in the location parameter, while the scale and shape parameters were held constant. For each simulated dataset, both a stationary GEV model and a non-stationary GEV model with a time-dependent location parameter were fitted using maximum likelihood estimation, and 100-year return levels were estimated.

Model	Mean	Bias	Standard deviation
Stationary	18.330	-2.257	2.838
Non - stationary	18.928	-1.660	6.144

Table 7: Performance measures for GEV models

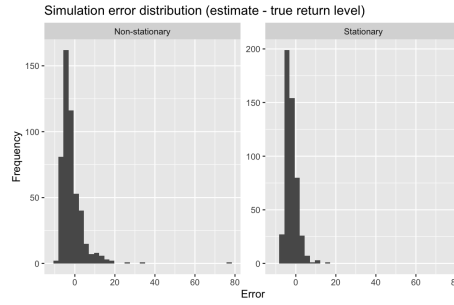


Figure 5: Error distribution in simulation

Table 7 summarizes the simulation results in terms of mean estimated return levels, bias, and standard deviation. The stationary model produced a mean estimate of 18.33 with a bias of -2.26 and a standard deviation of 2.84. In contrast, the non-stationary model yielded a slightly higher mean estimate of 18.93 and a smaller bias of -1.66, but with a substantially larger standard deviation of 6.14. This increased variability is also evident in Figure 7, which shows a wider and more right-skewed error distribution for the non-stationary model.

Overall, the simulation results illustrate a clear bias-variance trade-off. While the non-stationary model is better able to account for the underlying trend and therefore reduces bias in the estimated return levels, this improvement comes at the cost of considerably increased variability and higher overall estimation uncertainty. These findings suggest that non-stationary GEV models require larger sample sizes to achieve stable inference, whereas stationary models may provide more robust estimates when the available data are limited. Taken together, the inferential analysis indicates heavy-tailed behavior in extreme data breaches, no statistically significant temporal trend, and substantial uncertainty in long return period estimates.

5 Discussion and Conclusion

The purpose of this study is to investigate the behavior of extreme cyber security breaches using extreme value theory. By focusing on annual block maxima, the analysis specifically targets rare but potentially catastrophic data breaches, which are of particular relevance for risk assessment and decision-making. Implementing EVT methodology, the estimation of potential extreme events are suggested. The results show that the preferred model estimates a potential extreme data hack could arise to over 1 800 millions records lost for a once in fifty years event. When estimating an extreme event which potentially could occur once every 100 years the same model suggested over 2 700 million data records could be exposed. These findings does not contradict the results regarding the distribution in the data, which shows that annual maximum data breaches is heavy-tailed, as evidenced by a consistently positive shape parameter across both stationary and non-stationary models. This implies that extremely large cyber incidents, far exceeding those observed historically, remain plausible under the fitted models, meaning the estimates, whilst being much larger than any histrionically observed breach are not unrealistic. Still, the shape of the shape parameter may mean that risk assessments based solely on historical averages could substantially underestimate the potential severity of future breaches.

The empirical analysis does moreover not provide strong statistical evidence of a systematic upward trend in the magnitude of annual maximum data breaches. The estimated time trend in the non-stationary model was small and highly uncertain, and model comparison using AIC values and likelihood ratio tests favored the stationary model. This implies that, given the available data, the increase in model complexity does not translate into a better model fit, as the data does not show enough indication of a trend. Despite general concerns that cyber security risks may increase over time, such worries are not supported by the evidence in this analysis.

Estimates of return levels further highlights the limitations of extreme value modeling. While point estimates of the 50- and 100-year return levels suggest that future extreme breaches could involve several billion compromised records, the associated uncertainty is large. Using bootstrap resampling revealed a highly skewed distribution of return level estimates, which is mostly driven by sensitivity in the shape parameter and a limited number of block maximums. Although the median return level estimates are relatively stable, the upper bounds of confidence intervals are extremely wide, emphasizing the challenges of anticipating occurrences outside of the data set.

The simulation study provides important context for these findings. When data are generated from a process with a known positive trend, the non-stationary model reduce bias in return level estimation but at the cost of substantially increased variance. In contrast, the stationary model produce more stable estimates but does so at the cost of some bias when the true process was non-stationary. This illustrates a bias–variance trade-off which helps explain why

the non-stationary model was not favored in the empirical analysis. With short time series of annual maxima, the additional complexity of non-stationary models are simply not supported by data.

The findings in the report are subject to many limitations. It should be acknowledged that to constrain this report only one major extreme value modeling technique is used. The block maximum approach drastically reduces the number of observations, increasing uncertainty, and to fully explore the cyber security sector, a peaks-over-threshold approach would be the next step of the analysis. Furthermore, it is once again noted that the annual maximums are assumed to be independent from each other, which in practice very well may be violated as a consequence of evolving reporting practices or other structural changes in cyber security.

Conclusively, the analysis provides evidence of potential cyber attacks compromising billions of records. Whilst the results demonstrates the usefulness of extreme value theory when quantifying risk, the uncertainty in estimations points towards the struggles of producing stable estimations using block maximum approaches specifically on data regarding cyber security. Still, findings shows that the size of extreme data breaches mostly is characterized by heavy-tailed behavior and that there is no strong statistical evidence of a temporal trend over the observed period. The limited sample size when dividing data points into blocks rise uncertainty levels, and a stationary specification is preferred over a non-stationary alternative when fitting a model to the data. Future research would benefit from longer time series and alternative extreme value approaches such as the peaks-over-threshold method.

References

- [1] S. Coles, *An Introduction to Statistical Modeling of Extreme Values*. Springer-Verlag London, 2001.
- [2] E. Gilleland and R. Katz, “Extremes 2.0: An extreme value analysis package in r,” *Journal of Statistical Software*, 2016.
- [3] Z. Zhan, M. Xu, and S. Xu, “Predicting cyber attack rates with extreme values,” *IEEE Transactions on Information Forensics and Security*, vol. 10, no. 8, pp. 1666–1677, 2015. DOI: 10.1109/TIFS.2015.2422261.
- [4] K. Burnham and D. Anderson, “Model selection and multimodel inference,” *A Practical Information-theoretic Approach*, Jan. 2004. DOI: 10.1007/978-0-387-22456-5_5.
- [5] L. Lydon, “Corporate under reporting of cybercrime: Why does reporting to authorities matter?” M.S. thesis, Royal Holloway University of London, 2021.

6 Appendix

6.1 Model Diagnostics

```
fevd(x = max_millions, type = "GEV", method = "MLE")
```

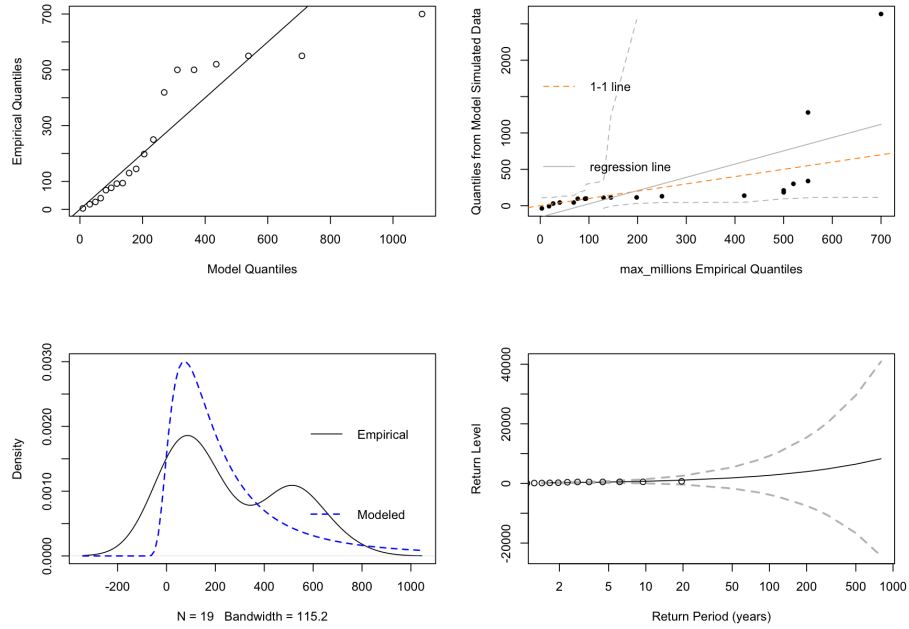


Figure 6: Stationary GEV diagnostics

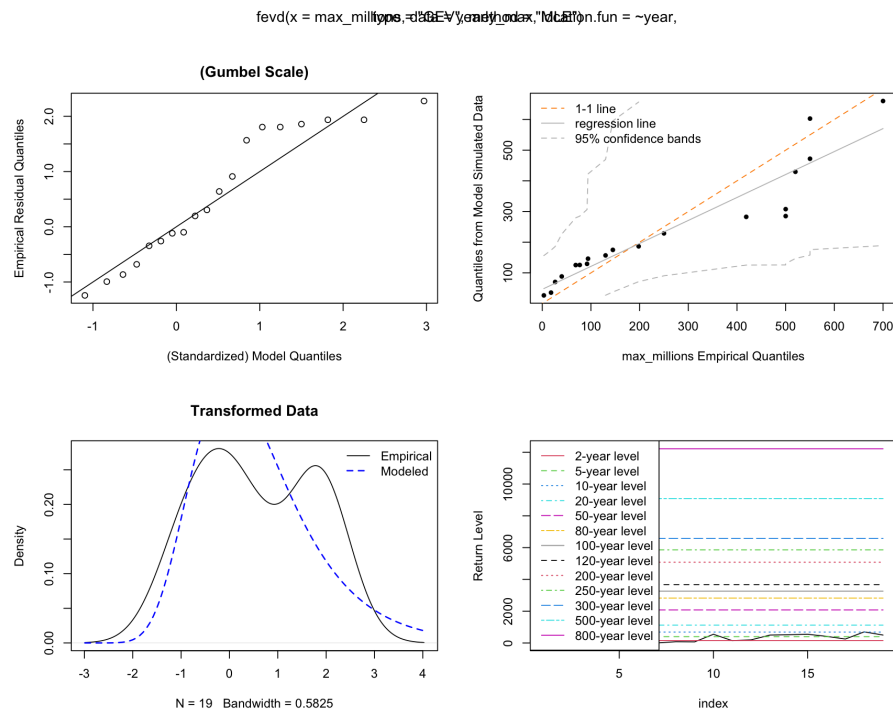


Figure 7: Non - stationary GEV diagnostics

6.2 Additional exploratory plots

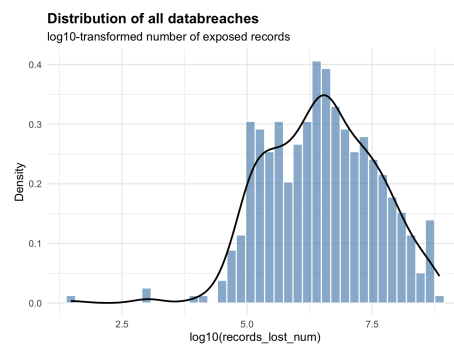


Figure 8: Distribution over all data breaches